Mini Project Report

Foundations of Programming-CSE10010

Topic-**Smart Traffic Signal Simulator**

Team members:

Om Jadhav – 1262252682

Niharika Singh – 1262253874

Yanika Phatak – 1262251867

Shriya Shelke – 1262252710

Project Guide:

Renuka Mane Ma'am

Institution Name:

MIT World Peace University, Pune

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- ## Acknowledgment

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- **Abstract**

Due to cities current rapid growth, traffic jams are becoming more common because there is an ever-increasing number of vehicles on the road. The traditional way that traffic signals currently operate is by using fixed-time intervals (aka "green lights") to allow vehicles to enter the intersection. Each fixed-time interval of a traffic signal does not accurately reflect (in real time) the actual density of traffic. The results are often longer wait times for vehicles at intersections (increased wait times), inefficient traffic flow (higher number of vehicles), and increased fuel consumption.

This project will develop and simulate a Smart Traffic Signal System using the C programming language, which is designed to determine which road will receive a green signal at a 4-way intersection based on how many vehicles are waiting at the intersection. During the simulation, input will be provided to the program to represent how many vehicles exist on each of the four roads and the number of vehicles waiting on each road will be used as the basis for comparison to determine which road has the highest density of vehicles. The road with the highest density of vehicles waiting at the intersection will receive priority and have its signal changed to the green signal. All other roads will remain on red.

This Smart Traffic Signal System will be created and simulated using the basic principles of programming, which include: variables, conditional statements, loops, and arrays. The above concepts will be used to create a logical decision-making system that simulates intelligent traffic control systems. The results of the simulation show how signals can be optimized based on traffic density. This project illustrates the application of programming to real-world problems while laying the groundwork for developing more sophisticated smart traffic control systems in the future.

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- ## Introduction

1. Problem statement:

Traffic congestion in urban areas, as well as other problems associated with an increase in number of vehicles, have been growing rapidly across the globe. Fixed time based traffic signal systems, commonly used in many regions, typically cycle through multiple roadways some successfully cycling through with little to no vehicles and others typically experiencing the opposite condition. This causes large amounts of waiting time for vehicles on high-volume roadways, creating queues, wasting fuel and increasing total travel times.

This project proposes to address this problem by creating an automated traffic signal control system that will determine which traffic signal should receive a green signal based on the number of vehicles at the intersection waiting for the signal. By providing more "green" time during peak traffic periods to the highest density roadway, it should result in improved traffic flow and a reduction in delays due to excessive wait times.

2. Background/Context:

Urban infrastructure relies heavily on effective traffic management systems. As more cities expand, and the use of automobiles continues to rise, the challenge is to effectively manage traffic. Current traffic signals are programmed based on a set standard timer with no regard to real-time traffic volume; therefore, they can create problems such as inefficient traffic control and congestion at intersections.

Smart and intelligent systems are being developed in the transition away from conventional traffic control systems. Technologies are combined sensors, cameras, data collection and processing and analysis software, etc. to develop smart traffic signal systems that measure traffic density and adjust the timing of traffic signals in conjunction with this collected data in real-time. In return, these systems provide a way to reduce wait times, enhance traffic flow, and make the transportation system much more efficient.

This mini project will demonstrate a simplified version of a traffic signal control system that is designed to mimic a smart traffic signal. The C programming language will be the platform of choice for programming the project, as opposed to using real-time sensors; vehicle counts will be entered into the control system; this will allow the control system to determine which road has the most vehicle traffic and allocate priority to that roadway.

3. Objective:

- The main objectives of this project are:
- To understand the concept of smart traffic signal systems.
- To simulate traffic signal decision-making using the C programming language.
- To develop a program that determines signal priority based on traffic density.
- To apply basic programming concepts such as variables, arrays, loops, and conditional statements.
- To demonstrate how programming logic can be used to solve real-world traffic management problems.

- **Literature Review**

Sr. No	Paper Title	Journal Name	Methodologies Used	Research Gap / Limitations	Performance Metrics
1	Traffic Management System Review	SAGE Journals	Review and classification of traffic management systems using data from multiple sources	Mainly theoretical review; lacks practical implementation details	Traffic efficiency and safety improvements
2	Fuzzy Logic Traffic Light Controller	Applied Soft Computing	Fuzzy logic based traffic signal controller with simulator comparison to conventional systems	Tested mostly in simulated environments	Reduction in waiting time at red lights
3	Adaptive Traffic Signal Control Using Sensors	Engineering Applications of Artificial Intelligence	Adaptive algorithms using sensor data and local communication between traffic lights	Requires sensor infrastructure and real-time data	Improved overall network traffic performance
4	SMARTS Microscopic Traffic Simulator	ACM Journal	Distributed microscopic traffic simulator supporting driver models and multi-vehicle simulations	High computational complexity for large systems	79.1% prediction accuracy and faster simulation speed
5	Fuzzy Intelligent Traffic Signal (FITS) Control	Control Engineering Practice	Embedded fuzzy logic control system integrated with signal hardware	Dependence on detectors and hardware reliability	Improved traffic mobility and flexible signal timing

6	Web-Based Traffic Light Simulation Software	Conference Paper	Browser-based traffic simulation using JavaScript and JSON-configurable road networks	Prototype level system with limited real-world validation	Real-time statistics of algorithm performance
7	Reinforcement Learning Based Adaptive Traffic Lights	IEEE	Reinforcement learning model trained using traffic simulation and real data	Requires extensive training data and computational resources	62.82% reduction in queue length and 56.37% reduction in waiting time
8	Smart City Traffic Congestion Management Simulator	IEEE	Real-time traffic visualization simulator with congestion optimization algorithm	Focuses mainly on simulation rather than deployment	Improved traffic flow and emergency vehicle prioritization
9	Traffic Signal Timing Optimization	Springer Nature	Optimization of traffic signal timing to reduce congestion at intersections	Limited adaptability to real-time traffic fluctuations	Improved traffic flow efficiency
10	IoT-Based Traffic Prediction Using OWENN Algorithm	Springer Nature	IoT data collection with OWENN neural network for traffic prediction and microprocessor-based signal control	Requires IoT infrastructure and datasets	98.23% prediction accuracy and 96.69% F score
11	Smart Traffic Signal Control Using Image Processing	International Journal of Computer Applications	Camera-based vehicle detection using image processing techniques	Performance affected by weather conditions and lighting	Vehicle detection accuracy and congestion reduction

12	IoT Based Smart Traffic Management System	International Journal of Engineering Research	IoT sensors and cloud-based monitoring to adjust signal timings dynamically	Requires large sensor network and stable internet connectivity	Reduced waiting time and improved traffic flow
13	Machine Learning Based Traffic Flow Prediction	Journal of Intelligent Transportation Systems	Machine learning models used to predict congestion and optimize signals	Requires large historical traffic datasets	Prediction accuracy and congestion reduction
14	Intelligent Transportation System for Urban Traffic	IEEE Conference	Sensor networks and communication systems for real-time traffic monitoring	High infrastructure cost and complexity	System response time and traffic throughput
15	Adaptive Traffic Signal Control Using Genetic Algorithm	Applied Soft Computing	Genetic algorithm used to optimize signal timing for intersections	High computational requirements for large networks	Reduction in vehicle delay and improved signal efficiency

TABLE 1 : Comparison of Research Papers in Literature Review

- ## System Design/Methodology

1. Overview of the Approach:

The Smart Traffic Signal Simulator is designed to demonstrate how traffic signals can dynamically adjust their timings based on traffic density. Traditional traffic signals operate on fixed timers, which often leads to unnecessary waiting even when some roads have very little traffic. To address this issue, the proposed system uses a simple decision-making logic implemented in the C programming language.

The simulator assumes a four-way traffic junction where each road has a certain number of vehicles waiting at the signal. The user manually enters the number of vehicles present on each road. Based on this input, the program compares the traffic density on all roads and determines which road should receive the green signal.

The system allocates the green signal to the road with the highest number of vehicles. The duration of the green signal can also be adjusted proportionally according to the number of vehicles present. This approach helps simulate the working principle of intelligent traffic management systems used in modern cities.

Although the system is a software simulation and does not interact with real traffic infrastructure, it effectively demonstrates how smart traffic control systems can improve traffic flow and reduce waiting time by making decisions based on real-time traffic conditions.

2. Block Diagram/Flowchart:

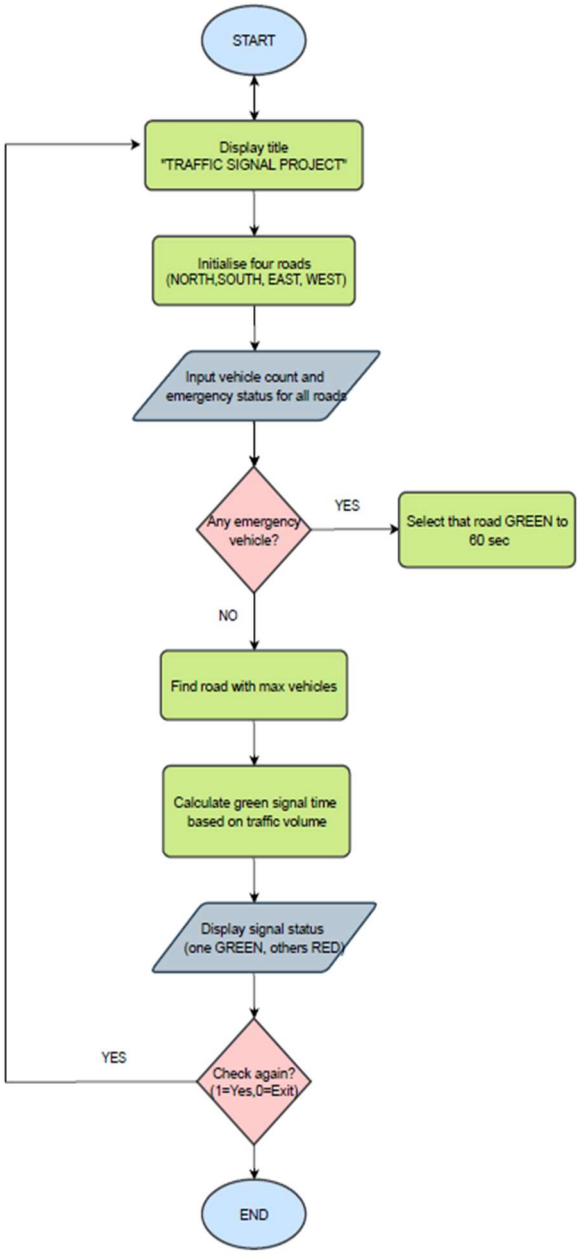

FIGURE 1 : Flowchart of Smart Traffic Signal Simulator

3. Components/Modules Used:

The Smart Traffic Signal Simulator is comprised of different modules enabling each section of the system to execute a specified function. The modular design offers an easier mechanism to construct, comprehend, and operate all modules within the system.

1. Input Module

The input module gathers traffic data from the user. In this instance, we are assuming a four-way intersection on which there are four separate roads: A, B, C and D. The user will specify the number of vehicles that occupy each road via the console interface. Each value is saved to respective program variables for processing later.

In practice, this data would come from sensors or cameras, not from the user entering it manu ally.

2. Processing Module

The processing module uses the data provided by the input module to process the vehicle counts on each road and derive the overall density of traffic. Using logical comparisons and basic programming structure created with C, this method efficiently and accurately completes the task, allowing it to determine which road has the most vehicle congestion.

3. Decision Module

In this module, the program will decide which road will receive the green signal. Conditional statements such as if–else are used to compare the number of vehicles on all roads so that the road with the highest number of vehicles is given priority, helping to reduce the amount of time spent waiting at the intersection.

4.Output Module

The output module displays the final outcome of the system. Once the program has determined which road will receive the green signal, it then displays this information in the console so that the user can see which road has been provided a green signal and which roads will remain on a red signal.

4. Algorithm:

Step 1: Start the program

Step 2: Initialize roads

Step 3: Begin infinite loop

Step 4: Take input

Step 5: Check for emergency vehicle

Step 6: Find maximum traffic

Step 7: Assign time based on traffic

Step 8: Display signal status

Step 9: Repeat or exit

- ## Implementation

1. Development Environment:

The Smart Traffic Signal Simulator was created in the programming language C. The implementation phase's goal was to turn the design for this smart traffic signal simulator into a functioning program that could model decision-making for traffic signals based on the density of vehicles using the intersection of the traffic signals.

The system was developed in the standard C Programming environment on a personal computer. The program was created, compiled, and executed using a C compiler. The console interface of the system allows the user to manually enter the number of vehicles on each of the roads leading to the traffic signal at the intersection.

The system assumes a four-way intersection of roads, with each road being represented by a variable (the number of vehicles on the road). The program compares the number of vehicles on the four roads and uses this to determine which of the roads should receive the green signal based on the density of vehicles on the road has the highest density of vehicles.

The implementation shows how to use basic C programming concepts; variables; conditional statements; loops; and functions. These concepts can be used to simulate the type of logic used in intelligent traffic signal systems.

Technologies Used:

- Programming Language – C
- Compiler – GCC / Turbo Compiler
- Operating System – Windows
- Interface Type – Console Based Simulation
- Libraries:
    1. stdio.h
    2. stdlib.h
    3. string.h

2. **Testing:**

The Smart Traffic Signal Simulator was tested to ensure it analyzes traffic density properly and assigns the green signal. To accomplish this, various scenarios with different numbers of vehicles on different roads were simulated. The output of the program was then analyzed to ensure that the road's density of vehicles would receive a green signal.

The system was also evaluated under simulated emergency scenarios, specifically involving the presence of emergency vehicles, such as ambulances or fire trucks, on a given roadway. Upon the simulator's detection of an emergency vehicle, the system would instantly grant a green signal to that vehicle, irrespective of the vehicular density on the remaining roadways. This particular testing methodology serves to validate the program's functionality under both emergency and standard traffic conditions.

Test Case	**Road A**	**Road B**	**Road C**	**Road D**	**Emergency Vehicle**	**Expected Output**
1	10	5	3	7	No	Green Signal → Road A
2	4	12	6	8	No	Green Signal → Road B
3	2	3	15	7	No	Green Signal → Road C
4	5	6	7	20	No	Green Signal → Road D
5	3	4	2	5	Road B	Green Signal → Road B (Emergency Priority)
6	8	6	4	2	Road C	Green Signal → Road C (Emergency Priority)

TABLE 2 : Test Cases and Testing Results of the System

The results of these test cases show that the system successfully identifies the road with the highest traffic density under normal conditions and also gives priority to emergency vehicles when present.

- **Results and Discussion**

1. **Output:**

The program processes input values that indicate the number of vehicles on four different roads that converge at the intersection of a traffic signal. Based on these inputs, the program calculates which road will have the green signal.

During regular traffic situations, the program compares the volume of vehicle traffic on each road to determine which road will be given the green signal according to the road that has the largest volume of traffic. This allows vehicles waiting at intersections that have a lot of traffic congestion to have less wait time and overall provide improved traffic flow throughout the intersection.

In addition to normal traffic scenarios, the program also takes into account the presence of emergency vehicles, which include ambulances, fire trucks, or police cars.

```
 SMART TRAFFIC SIGNAL SIMULATOR (DENSITY)

Road: NORTH
Enter vehicle count: 5
Is there an emergency vehicle? (1 for Yes, 0 for No): 0

Road: EAST
Enter vehicle count: 2
Is there an emergency vehicle? (1 for Yes, 0 for No): 0

Road: SOUTH
Enter vehicle count: 6
Is there an emergency vehicle? (1 for Yes, 0 for No): 1

Road: WEST
Enter vehicle count: 3
Is there an emergency vehicle? (1 for Yes, 0 for No): 0


          SIGNAL STATUS REPORT
 [!] EMERGENCY DETECTED ON SOUTH ROAD [!]

 [NORTH]       Status: RED (STOP)                    (Vehicles: 5)
 [EAST]        Status: RED (STOP)                    (Vehicles: 2)
 [SOUTH]       Status: GREEN (GO)      Time: 60 sec  (Vehicles: 6)
 [WEST]        Status: RED (STOP)                    (Vehicles: 3)


Do you want to simulate another cycle?
1. Yes
0. Exit
Enter choice: 0
System shutting down...
[1] + Done                       "/usr/bin/gdb" --interpreter=mi --tty=${DbgTerm} 0<"/tmp/Microsoft-M
IEngine-In-qo15ud2x.wql" 1>"/tmp/Microsoft-MIEngine-Out-51atxd3r.ks2"
```

FIGURE 2: Program Output of Smart Traffic Signal System

FIGURE 3 :Graphical Representation of Traffic Signal Output

2. Analysis:

Simulation results indicate that the system is operating per its intended logic. The program can evaluate normal traffic flow conditions to find which route has the heaviest traffic density and provide that route an active green light.

When an emergency vehicle is detected, the system will ignore normal traffic flow rules and provide an active green light to the route that contains the emergency vehicle. This allows the emergency response team to move quickly and improve the overall effectiveness of the traffic control system.

These results show that combining emergency vehicle priority with traffic density analysis can improve traffic flow management at intersections. Although the current system is a rudimentary simulation of how intelligent traffic signal systems function in modern smart cities, it provides a clear representation of the working principles of intelligent traffic signal systems that are used today.

- ## Conclusion

The Smart Traffic Signal Simulator project was created to display how we can optimize operation of traffic signals by factoring in the density of vehicles on the roadway rather than simply utilizing fixed time intervals. Historically traffic signals have been based on fixed timers and can result in significant delays due to high volumes of traffic on one direction of the street compared to very low volumes of traffic on another direction of the street.

This project's purpose was to create a simple decision-making process for traffic signals based upon the four streets of the intersection and to program the solution using C++. This means that the program uses the number of vehicles on each of the four roads to come up with an output based on the density of each road. The output will assign the road with the highest number of vehicles to receive the green light based on available time to get the vehicles through the intersection.

The test results demonstrate that the system can determine the highest-volume traffic road and allocate a green light to that road; therefore, the overall project illustrates the function of "smart" traffic-signal control and merges basic programming logic into a computer-based operation for real-world traffic control systems.
Finally, the simulator will also allow for emergency vehicles (like ambulances or fire trucks) to have automatic priority in order to get them through intersections quicker than typical traffic.

- **Future Scope:**

The Smart Traffic Signal Simulator that was made for this project shows how traffic signals can be controlled based on how many vehiclesre on the roads nearby and if there are emergency vehicles that need to go first. It has shown that it can work like a Smart Traffic Signal System but there are many ways to make it even better.

For example the system would be a lot better if it had Real-Time Traffic Monitoring technologies like sensors, cameras or IoT devices. These devices can count how many vehicles are on each road and send that information to the traffic control system. This way the Smart Traffic Signal System can change the timing based on what is really happening on the roads not just what it thinks will happen.

Another good thing to add would be to use Artificial Intelligence or machine learning to look at data and predict when and where there will be traffic jams. The system can use this data to figure out how traffic usually moves at different times of day on different days and in different weather so it can get ready for traffic jams and make the signal timing better.

The simulator could also be made better by creating a user interface that uses graphics to show what is happening with all the signals, vehicles and their positions at intersections.

This project could also be made bigger by making Smart Traffic Signal Systems that can control intersections, not just one. All of these changes will make the Smart Traffic Control Simulator a better smart traffic management solution than anything that exists now.

The Smart Traffic Signal Simulator will be a lot better with these changes. The Smart Traffic Signal System will be able to control traffic signals in a way. The Smart Traffic Signal System will make traffic move smoothly. The Smart Traffic Signal System will be a good thing, for people who drive.

- **Research Paper/Patent Details**

Currently, no academic articles or patents have been produced regarding the Smart Traffic Signal Simulator that was built through this project. This project was completed as part of a mini project with the goal of initially showing how existing systems could be made better by implementing simple decision making logic using the traffic density of an area.

This project has been created with an educational focus on applying the programming concepts of using Variables, Conditional Statements, Loops and Logical Comparisons through the use of the C Programming Language. Therefore the final product will not yet have been officially sent to a research journal, conference or patent office.

However, there is an opportunity for this project to be developed into further advanced project. All that is required are upgrades such as incorporating sensory devices or cameras to provide real-time monitoring of vehicle density, combining the project with IoT (Internet of Things) technologies, and implementing machine learning algorithms to gain insight into potential future vehicle densities.

If future upgrades to the Smart Traffic Signal Simulator are successful, then it may be appropriate to create a knowledge document and submit it to an academic journal that covers Intelligent Transportation Systems or Smart City Technologies. Since the project is currently still considered to be a proof of concept from the standpoint of research, and it does not currently have any formal publication or patent application.

- **Plagiarism Check**

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- **Appendix**

The appendix will provide additional supportive materials regarding the Smart Traffic Signal Simulator project. These materials will offer greater insight into how the system was implemented, thereby providing readers with details on how to operate the program.

The appendix will consist of the total source code for the Smart Traffic Signal Simulator written in C. Having this coding information allows readers to see the logic used by comparison of traffic density, allocation of signals and handling of emergency vehicles. In addition, this coding provides examples of how programming concepts (i.e., variables, conditional statements, loops and functions) were utilized in a practical manner.

```c
    strcpy(roads[2].name, "SOUTH");
    strcpy(roads[3].name, "WEST");

    do {
        clearScreen();
        printf(" SMART TRAFFIC SIGNAL SIMULATOR (DENSITY) \n");

        // 1. Get Input
        getTrafficData(roads);

        // 2. Logic: Decide which road gets Green
        int greenIndex = findGreenSignalIndex(roads);

        // 3. Logic: Calculate time duration
        //Emergency vehicles get max time by default or based on density
        int duration = calculateDuration(roads[greenIndex].vehicleCount);
        if(roads[greenIndex].isEmergency) {
            duration = TIME_HEAVY; // Give max time for emergency to clear
        }

        // 4. Output: Show the signals
        displaySignals(roads, greenIndex, duration);

        // Menu Loop
        printf("\nDo you want to simulate another cycle?\n");
        printf("1. Yes\n");
        printf("0. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);
```

```c

    } while (choice != 0);

    printf("System shutting down...\n");
    return 0;
}

//Function Definitions

// Function to clear console
void clearScreen() {
    #ifdef _WIN32
        system("cls");
    #else
        system("clear");
    #endif
}

// Function to take user input for all 4 roads
void getTrafficData(Road roads[]) {
    int i, emergInput;
    for (i = 0; i < 4; i++) {
        printf("\nRoad: %s \n", roads[i].name);

        printf("Enter vehicle count: ");
        scanf("%d", &roads[i].vehicleCount);

        // Validate negative input
        if(roads[i].vehicleCount < 0) roads[i].vehicleCount = 0;

```

```c

        printf("Is there an emergency vehicle? (1 for Yes, 0 for No): ");
        scanf("%d", &emergInput);
        roads[i].isEmergency = (emergInput == 1) ? 1 : 0;
    }
}

// Function to determine which road gets the Green Light
// Priority: Emergency > Highest Traffic > First Road (in case of tie)
int findGreenSignalIndex(Road roads[]) {
    int i;
    int maxTrafficIndex = 0;
    int maxTraffic = -1;

    // 1. Check for Emergency Vehicles first (Highest Priority)
    for(i = 0; i < 4; i++) {
        if(roads[i].isEmergency == 1) {
            return i; // Immediate priority to this road
        }
    }

    // 2. If no emergency, find road with Maximum Traffic
    for (i = 0; i < 4; i++) {
        if (roads[i].vehicleCount > maxTraffic) {
            maxTraffic = roads[i].vehicleCount;
            maxTrafficIndex = i;
        }
    }

    return maxTrafficIndex;
```

```c
}

// Function to calculate green light duration based on density
int calculateDuration(int vehicleCount) {
    if (vehicleCount > 20) {
        return TIME_HEAVY;     // 60 seconds
    } else if (vehicleCount >= 10) {
        return TIME_MODERATE;  // 40 seconds
    } else {
        return TIME_LIGHT;     // 20 seconds
    }
}

// Function to display the final status of the junction
void displaySignals(Road roads[], int greenIndex, int duration) {
    printf("\n\n");
    printf("           SIGNAL STATUS REPORT            \n");

    if (roads[greenIndex].isEmergency) {
        printf(" [!] EMERGENCY DETECTED ON %s ROAD [!]\n\n", roads[greenIndex].name);
    }

    for (int i = 0; i < 4; i++) {
        if (i == greenIndex) {
            // This road is Green
            printf(" [%s] \tStatus: GREEN (GO) \tTime: %d sec \t(Vehicles: %d)\n",
                   roads[i].name, duration, roads[i].vehicleCount);
        } else {
            // All other roads are Red
            printf(" [%s] \tStatus: RED (STOP) \t\t\t(Vehicles: %d)\n",
                   roads[i].name, roads[i].vehicleCount);
```

```c
        }
    }
    printf("\n");
}
```